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TASK 01 – UK SPONSORSHIP VISA FOR EMPLOYERS

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1. Introduction

The UK operates a sponsorship visa system to assist the arrival of people from overseas, largely for university study or to take up work opportunities. In order to recruit international students or workers, universities and employers must be licensed sponsors, and they provide the necessary documentation required to apply for visas. For study entrants, the key document is the CAS; for those entering to work, it's the CoS.

The ways in which individuals relocate to the UK have changed greatly in recent years. Yes, Brexit brought with it new immigration rules, but all of the changes do not stop there. There has been unrest in the global economy, staffing shortages in many industries, and a few aftershocks of the COVID-19 pandemic are still in play. Taken together, all these events have provided significant insight into how universities and businesses sponsor people moving to the UK. Therefore, sponsorship data is especially critical right now.

This project aims to reveal what is really going on with the UK sponsorships. We are using SQL Server to organize and clean the historical data, then load it into Microsoft Power BI to build visualizations and detect patterns. The answers to the questions would include the following: what are the long-term trends, how do things shift quarter to quarter, which countries are most represented, and how do work sponsorships stack up against student sponsorships?

2. Data Source & Extraction

The data we're using comes straight from the UK government. We downloaded it from Data.gov.uk, which is where they publish official open data. There are four main datasets that cover both study and work sponsorships:

Study Sponsorship Data

- **CAS_D01**: Shows study sponsorship applications broken down by year and quarter, looking at things like what type of application it was and what kind of institution issued it
- **CAS_D02**: Similar to the first one, but also tells us where students are coming from geographically and their nationalities

Work Sponsorship Data

- **CoS_D01**: Tracks work sponsorship applications by year and quarter, showing the type of application, visa category, and which industry the job is in
- **CoS_D02**: Also tracks work sponsorships by year and quarter, but includes geographic regions and nationalities of the applicants

3. SQL Server Implementation

A separate SQL Server database was created for storing and managing sponsorship data and it was named Uk_Immigration. Using the Import Flat File in SQL Server, the CSV files were imported into SQL Server, with each data set linked to its respective table.

3.1 Data Cleaning & Transformation

One of the major issues that were found to affect data quality while importing the data was that the applications column in the four tables was represented by the data type of NVARCHAR, since it contains comma-separated values like "1,234." This is a major issue since numerical analysis is carried out using integer values. We removed the comma and then changed the data type to int.

```
-- Removing commas and turning updationg application column data type in all tables

UPDATE CAS_D01
SET Applications = REPLACE(Applications, ',', '');
ALTER TABLE CAS_D01
ALTER COLUMN Applications INT;

UPDATE CAS_D02
SET Applications = REPLACE(Applications, ',', '');
ALTER TABLE CAS_D02
ALTER COLUMN Applications INT;

UPDATE CoS_D01
SET Applications = REPLACE(Applications, ',', '');
ALTER TABLE CoS_D01
ALTER COLUMN Applications INT;

UPDATE CoS_D02
SET Applications = REPLACE(Applications, ',', '');
ALTER TABLE CoS_D02
ALTER COLUMN Applications INT;

-- remove blanks in application column

UPDATE CAS_D01 SET Applications = NULL WHERE Applications = '';
UPDATE CAS_D02 SET Applications = NULL WHERE Applications = '';
UPDATE CoS_D01 SET Applications = NULL WHERE Applications = '';
UPDATE CoS_D02 SET Applications = NULL WHERE Applications = '';
```

To achieve proper chronological sorting and mathematical calculations through time, the numeric ordering of quarters became relevant. This involved using a calculated quarter number to signify the order of the quarters through the years instead of using text values for the quarters.

```sql
-- Clean Quarter column (extract quarter number)

ALTER TABLE CAS_D01
ADD QuarterNumber AS CAST(RIGHT(Quarter, 1) AS INT) PERSISTED;

ALTER TABLE CAS_D02
ADD QuarterNumber AS CAST(RIGHT(Quarter, 1) AS INT) PERSISTED;

ALTER TABLE CoS_D01
ADD QuarterNumber AS CAST(RIGHT(Quarter, 1) AS INT) PERSISTED;

ALTER TABLE CoS_D02
ADD QuarterNumber AS CAST(RIGHT(Quarter, 1) AS INT) PERSISTED;


-- Create QuarterSort column

ALTER TABLE CAS_D01
ADD QuarterSort AS (Year * 10 + CAST(RIGHT(Quarter, 1) AS INT)) PERSISTED;

ALTER TABLE CAS_D02
ADD QuarterSort AS (Year * 10 + CAST(RIGHT(Quarter, 1) AS INT)) PERSISTED;

ALTER TABLE CoS_D01
ADD QuarterSort AS (Year * 10 + CAST(RIGHT(Quarter, 1) AS INT)) PERSISTED;

ALTER TABLE CoS_D02
ADD QuarterSort AS (Year * 10 + CAST(RIGHT(Quarter, 1) AS INT)) PERSISTED;
```

After that all the data columns were standardized to get the same format throughout all the tables.

Then to make it easier to report, a unified SQL view was developed for study sponsorships and work sponsorships. This is achieved by using the UNION ALL command to combine data from the respective D01 or D02 table. It should be noted that the SQL view is not designed to remove null values but to provide a standardized fact structure for easier interpretation by Power BI. Notably, the nulls were retained in the consolidated data views since they represent real differences in the data structure between datasets, as opposed to missing or incorrect data.

```sql
CREATE OR ALTER VIEW Study_Sponsorship AS
SELECT
     Year,
     Quarter,
     QuarterNumber,
     QuarterSort,
     Type_of_application,
     Institution_type_group,
     Institution_type,
     NULL AS Geographical_region,
     NULL AS Nationality,
     Applications
FROM CAS_D01

UNION ALL

SELECT
     Year,
     Quarter,
     QuarterNumber,
     QuarterSort,
     Type_of_application,
     Institution_type_group,
     NULL AS Institution_type,
     Geographical_region,
     Nationality,
     Applications
FROM CAS_D02;
```

4. Power BI data connection and Transformation

The tables were imported directly to power bi by connecting them and we chose the 2 unified views tables for our analysis

Only light transformations were performed with Power BI since majority of data cleaning was carried out in SQL Server. The major transformations with Power BI involved,

Verifying data types

Renaming columns

Creating calculated measures using DAX –

Total Applications =

[Total Study Applications] + [Total Work Applications]

Total Study Applications =

SUM (Fact_Study[Applications])

Total Work Applications =

SUM (Fact_Work[Applications])

Study to Work Ratio =

DIVIDE ([Total Study Applications], [Total Work Applications])

Study % Share =

DIVIDE ([Total Study Applications], [Total Applications])

Work % Share =

DIVIDE ([Total Work Applications], [Total Applications])

These are some of the DAX measures that were calculated for more precise and important visualizations.

Notably, the null transformations were not removed. This was done differently, where nulls were managed through graph-level filters for precise analysis.

5. Data Model & Relationships

In the Power BI data model, a star-schema setup was used, with Unified study sponsorship data and Unified work sponsorship data being the main fact tables. Date, Nationality, Geographical Region, Institution Type, Industry, and Leave Category all belonged to the dimension side.

One-to-many relationships were established between the fact table and the dimension tables where applicable. Each attribute in the dimension tables possessed unique values while the fact table held many records linked to those attributes. Cross-filtering was kept mostly one-way to avoid confusion. We avoided many-to-many links due to the complexity they add and how it can skew aggregation results. Several of the attributes did not make it into the dimension tables as they had either too many nulls or duplicates. Rather than force artificial connections, we evaluated those attributes at the fact level in visuals.

6. Dashboard Visualization Analysis

6.1 Overview

- The total applications card shows combined total volume of Study and Work sponsorship applications reached 16 million. This large volume underscores the UK's significance as a destination for international students and workers.

- Study to work ratio card represents that for every 1.74 Study applications, there is 1 Work application. This indicates that the Study stream (CAS) is significantly larger than the Work stream (CoS), which is typical for global migration trends.
- Latest quarter applications card shows a volume of 328,000 applications in the latest quarter shows strong, sustained demand for sponsored visas.
- Total study and work applications by year line chart shows both Study and Work applications show a dramatic and steep upward trend, particularly starting around 2020. Study applications consistently outpaced Work applications. The line for Total Study Applications is significantly higher over the years. Clear increase post-2020 suggests a strong, positive response to recent UK immigration policy changes designed to attract students and skilled workers.
- Total applications by geographical regions shows south Asia leads the applications with around 2 million applications followed by east Asia and sub Saharan Africa.
- Study and Work QoQ growth by year represents both Study QoQ Growth and Work QoQ Growth show high volatility across the years, but with overall high growth rates visible by 2020. Growth rate frequently surpasses 100% and reaches nearly 400% in some periods for both categories, indicating very rapid expansion. The high growth percentages (rather than volume) confirm that recent years have seen an exponential increase in demand, not just a gradual rise.

6.2 Study Sponsorship

- Total study applications has 10M applications confirming the Study stream is the largest component of the overall 16M applications.
- Study % share card shows study applications account for 64% of the total combined application pool.
- Study QoQ growth card show the latest Quarter-over-Quarter growth rate is exceptionally high at 1692.82%, emphasizing the explosive demand in the most recent periods.
- Top Study Applications by Nationality bar chart represent top 3 Nationalities where China leads significantly with 1,614,458 applications. India follows with 782,071 applications. Nigeria ranks third with 265,997 applications. The Top 3 countries (China, India, Nigeria) account for the vast majority of sponsored study visa demand, a key finding for the "Top 10 Nationalities" requirement.
- Total Study Applications by Institution type group bar chart shows the 'Higher Education' group (Universities) is overwhelmingly dominant. This group accounts for the bulk of the 10M total study applications, confirming the primary attraction of the UK's university sector. The 'All other institutions' group shows a smaller, but still significant, volume, representing demand across colleges, schools, and private providers.
- Total Study Applications by Institution type shows 'Non Russell Group' and 'Russell Group' institutions have the highest volumes, consistent with the dominance of the university sector. Categories like 'Tertiary, Further Edu' show the next highest volumes, confirming that application volume is spread across different educational levels.

6.3 Work Sponsorship

- Top work applications card confirms the Work stream's volume, making up the remaining 6M of the overall 16M applications.
- Work % share card shows work applications account for 36% of the total combined application pool.
- Work QoQ growth shows the latest Quarter-over-Quarter growth for the Work stream is extremely high at 2728.96%. This suggests that work migration policies have seen a massive, recent surge in utilization.
- Total Work Applications by Nationality shows us the top 3 Nationalities. India leads the work sponsorship demand with 872,585 applications. United States ranks second with 394,420 applications. Nigeria ranks third with 170,635 applications. Unlike the Study stream (led by China), the Work stream is led by India, followed by applicants from the United States, indicating a strong demand for skilled labor from North America.
- Total Work Applications by Industry represent leading Industries - 'Human Health and Social Work' and 'Arts, Entertainment, and Recreation' show the highest volumes of work applications. Other Major Sectors - 'Information and Communication' and 'Professional, Scientific, and Technical Activities' also represent major industry users of the sponsorship system, confirming the demand for highly skilled labor.
- Total Work Applications by Category of Leave show dominance of Skilled Worker: 'Worker (Previous Tier 2)' visa category (the Skilled Worker route) is the overwhelming category for work applications, with volumes significantly higher than 2M. Temporary vs. Skilled: The 'Temporary' worker category has a much lower volume, indicating that the vast majority of sponsored work applications are for long-term, skilled employment, rather than short-term placements.

7. Key Findings & Insights

7.1 Study sponsorships

Another important insight is the clear lead of study sponsorship applications over work sponsorships throughout the entire timeframe analyzed. The Overview dashboard reveals the presence of study-related applications in approximately two-thirds of sponsorship total activity; this aspect is further emphasized by the Study % Share KPI and Study to Work Ratio at approximately 1.7.

This materializes the fact that international education plays a crucial part in migration to the UK. The continuity of study sponsorship applications implies that this phenomenon may not be merely short-term but structural in nature as well. Education remains at the heart of UK migration and economic policy because UK universities retain a substantial number of international students.

7.2 QoQ Dynamics

When analyzing the discharge on quarter-over-quarter changes, there are various instances showing significant fluctuations in study sponsorships. In certain periods, the overall quarter-over-quarter growth increases significantly beyond 1000%.

This might seem counterintuitive at the outset, but this is merely the mathematical revelation of the very low base established by the previous quarter's values. Volatility is also exhibited by the Work sponsorships category on a quarter-over-quarter basis, but the variations are relatively less extreme than those identified for study sponsorships.

This is because the respective work visas are influenced by the fluctuations within the labor market and the overall economic trends on the respective sound foundations of time frames of change.

7.3 Nationality Concentration

When analyzing the pattern of sponsorships based on nationality both in the context of education and employment, it becomes apparent that there is a massive flow of applicants coming from a few select nations. The primary sources of sponsorships in education include the USA, India, Nigeria, China, and Pakistan. On the other hand, the primary sources of sponsorships in employment include India, along with a few other nations with significantly lower volumes.

The implication is that the dependence on these nations could have a destabilizing effect on the flow of sponsorships in the future due to geopolitical or other factors. The point to note is that the stability that comes with having multiple sources will provide an edge in the long term.

7.4 Institutional and Sectoral Dependence

As indicated by the sponsorship dashboard, top universities are driven by the issuance of sponsorships. The vast majority of applications for studies and work sponsorships relate to non-Russell Group and Russell Group universities, indicating that internationally, the UK has an extremely strong position regarding education.

In contrast, work sponsorships show a greater clumping by industry. The health, information technology, professional, and associated industries drive work sponsorships, and of course, there is a heavy emphasis on health care, which can very well be attributed to skill shortages, an aging population, and deficiencies in the makeup of the workforce in the UK. A category of leave analysis indicates that a few channels are generating volumes of work.

7.5 Regional Patterns of Sponsorship Demand

Regionally, sponsorship for research and employment comes primarily from Asia, and within this region, it comes from South and East Asia. A large volume also comes from Africa, and a smaller volume comes from the Americas, Europe (apart from the UK), and the Middle East. The regional

pattern reflected in the dashboards represents the international relationships that exist for and from the UK and indicates disparities that might affect future migration policies and interactions.

8. Conclusion

In this report, a walkthrough is performed on an entire data analysis process, which involves SQL Server and Power BI for analyzing support sponsorship visas that are issued in the United Kingdom. SQL Server is a solid platform for data analysis and integration that guarantees data integrity from one analysis phase to another.

Lastly, the application of Power BI is utilized for the development of a complex data model with the correct linkages. This engenders the development of a multi-page dashboard that comprises broad executive summaries and other analyses. This is covered within the long-term trends, recent variations, nationality clusters, industry dependency, and sponsorship application geographic diffusion.

Analysis of the results shows that the biggest form of sponsorships leading the flow of sponsored migrations in the UK is sponsorships through studies, although work sponsorships continue to play an important albeit more fluctuating role, as one would expect influenced by the state of the labor market. The results highlight areas of both strength and weakness in the sponsor system in terms of sponsor sources and industries.